

Imanol Gómez

Technical lead & AI engineer

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PROFILE

Technical lead and AI engineer with 15+ years of building production software end to end in Python and C++, in automotive AI, medical devices and real-time embedded systems. As Head of AI Deployment at CARIAD (Volkswagen Group), I lead 12 engineers taking AI models from research prototypes to production on a volume-production ADAS platform, working directly with Volkswagen Group brands and with Qualcomm, Samsung and NVIDIA. Before that I built an AI tele-monitoring medical product from requirements to production under IEC 60601, and as a creative technologist I delivered installations, taught, gave talks and ran workshops. In my own time I build LLM agents with automated evaluation and on-device AI on NVIDIA Jetson.

SKILLS

LLM & agents	Agentic workflows (LangGraph), RAG & pgvector, MCP, Structured outputs (Pydantic), Prompt engineering, LiteLLM / OpenRouter
LLM inference & serving	vLLM, SGLang, TensorRT-LLM, SLO capacity benchmarking (TTFT, ITL, NVIDIA AIPerf), Prefix / KV-cache reuse, Quantization (AWQ, GPTQ, TensorRT Model Optimizer), GPU cost per task
Machine learning	PyTorch, TensorFlow, Fine-tuning, ONNX, TensorRT, Vision-language driving planner (EMMA-VLM), Model optimization & deployment
Software engineering	Python (production), C++, Rust, FastAPI, PostgreSQL, Docker & Compose, GitHub Actions / CI-CD, HTMX / Jinja, Linux & snap packaging
Edge & embedded	Qualcomm NPU, NVIDIA Jetson Orin, ROS 2, QNX deployment, Embedded Linux, RTOS & bare-metal MCU firmware, MCUs: Nordic nRF53, ESP32, STM32, Raspberry Pi Pico, Real-time GPU applications (30+ FPS)
GPU & parallel computing	CUDA C++, cuTile Rust & cuda-oxide, GPU profiling (Nsight Compute & Systems), CUDA/OpenGL interop, OpenCL, GLSL, wgpu / WGSL compute shaders (Rust), Real-time GPU particle and flocking simulations, TensorRT inference on NVIDIA Jetson
Regulated environments	IEC 60601 (healthcare), ISO 26262 (automotive), ASPICE, Compliance & auditability
Delivery & leadership	Team leadership (12 engineers), Product ownership (SAFe PO/PM), Stakeholder management, Mentoring, Talks & workshops

PROFESSIONAL EXPERIENCE

Oct 2025 – Present
Berlin

CARIAD

Head of AI Deployment Case study

Leads a team of 12 engineers that takes AI models from research prototypes to production on a volume-production ADAS platform (CARIAD–Bosch Automated Driving Alliance).

- Owns production reliability, performance and auditability under ISO 26262 and ASPICE: latency, memory and power targets, release quality and traceability.
- Works directly with enterprise stakeholders at Volkswagen Group brands (Volkswagen, Audi) on requirements, priorities and delivery.
- Leads technical discussions with semiconductor partners Qualcomm, Samsung and NVIDIA on integration and performance.

Apr 2023 – Oct 2025
Berlin

CARIAD

Senior ML Software Engineer Case study

Technical Product Owner for the AI Runtime Environment; developed and deployed AI perception models on Qualcomm accelerators.

- Built a Python SDK for model conversion, quantization and optimization, plus a runtime abstraction layer used by engineering teams across the organization.
- Supported internal customer teams end to end, from scoping their use cases to production integration.
- Developed sensor-fusion, 3D-occupancy and behaviour-prediction models and deployed them on Qualcomm AI accelerators under real-time constraints.

Aug 2020 – Apr 2023
Berlin

NELA by Carepath Technologies

AI Software Engineer

Built NELA Cough, an AI-based tele-monitoring product, in a regulated healthcare environment.

- Developed the on-device symptom monitoring, in line with IEC 60601: cough detection in continuous audio with speech removed, plus ambient light, temperature, humidity and air-quality sensing.
- Led the IoT system architecture end to end, from requirements to production: Azure IoT cloud back end and device firmware.
- Worked with external medical hardware suppliers through to series production: 250 devices produced and shipped to 50 patients for clinical trials by April 2023.

Jun 2014 – Aug 2020
Berlin & international

Self-employed

Creative Technologist Case study

Scoped and delivered client projects end to end: interactive installations, light and video projection, live performance software and fashion tech.

- ProtoPixel (2017 – 2020): firmware and software for real-time LED lighting products; delivered the main product software as a snap package; technical evangelist.
- ElektroCouture (2016 – 2018): led design and development of wearable products; main developer of the wearable driver firmware; fashion-tech mentor.
- Engineered and ran live the show software for the multimedia dance performance Murmur (Aakash Odedra Company) on its world tour.

Oct 2011 – Jul 2014
Linz

Ars Electronica Linz

Creative Engineer

Built interactive computer-vision installations in C++, Python, OpenGL and OpenCV.

- Lead developer of GeoPulse, a spatial-data visualization platform (Ars Electronica Center).
- Techkriti 2012 "Brain Battle", MED-EL Audioversum exhibits.

PROJECTS

- **Alibi: LLM agents that lie** github.com/ImanolGo/alibi
  Python, LangGraph, pgvector, MCP, FastAPI, LiteLLM, OpenTelemetry
  A murder-mystery game where every suspect is an LLM agent — with RAG memory, tool use and eval gates in CI. 2.9% contradiction rate across evaluation runs, at $0.03 per run.
- **How many AI agents fit on one GPU?** github.com/ImanolGo/agents-per-gpu/blob/main/reports/REPORT.md
  Python, vLLM, SGLang, TensorRT-LLM, AIPerf
  A benchmark of vLLM, SGLang and TensorRT-LLM on agent-shaped traffic: concurrent agents served within a latency SLO, and the cost per agent task. Prefix caching took one A40 from 2–3 to 6 concurrent agents, at a third to two fifths of the cost per task
- **A fully on-device AI robot assistant** github.com/ImanolGo/rover-assistant
  Python, ROS 2, TensorRT, DeepStream, Whisper, YOLO, llama.cpp
  Voice and vision on an NVIDIA Jetson Orin Nano — nothing leaves the robot. Vision answers in ~37 ms and the language model runs at ~30 tokens/s, all on the robot.
- **CUDA particles: Particle Atlas** github.com/ImanolGo/cuda-particles
  CUDA, C++, OpenGL, GLSL, GLFW
  Learning CUDA through six small lessons and an interactive particle artwork with glowing trails.

- **One starling flock on the GPU, four ways** github.com/ImanolGo/murmuration-kernels/blob/main/docs/ARTICLE.md
  CUDA C++, cuTile Rust, cuda-oxide, wgpu, Rust, Nsight
  The same flock step written in CUDA C++, cuda-oxide, wgpu and cuTile Rust, checked against a CPU reference and profiled with Nsight. Same flock, four GPU stacks, one CPU reference, Nsight profiles on two GPUs

Open source: contributor to openFrameworks (C++); 80+ public repositories on GitHub.

Talks & workshops: Llum BCN Festival · Factory Berlin · Porto Tech Hub · iSpace Accra · Goethe-Institut Bangkok · Scope Sessions Berlin; university teaching at Universitat Pompeu Fabra and HfG Offenbach.

EDUCATION

2009 – 2011
Barcelona

Master's Degree, Sound and Music Computing
Universitat Pompeu Fabra
Focus: computational modeling, perception, cognition and interactive systems.

2003 – 2009
Madrid

Master's Degree, Telecommunication Engineering
Universidad Autónoma de Madrid
Specialization: Multimedia Signal Processing, Electronic Systems.

- Erasmus year at TU Berlin (2007 – 2008).
- Research student at Fraunhofer HHI, Image Processing Department (2008 – 2009): forward error correction for video compression.

LANGUAGES

Spanish Native

English Professional proficiency

German Professional proficiency